

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1751

Yabucoa 100 kV
September 17, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1751. The Interconnection Customer proposes a 165.4 MW solar photovoltaic facility interconnecting at the Yabucoa 100 kV substation in the Illinois area. The request was received on May 18, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 536 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,119,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: YABUCOA - SATICOY 100.0 kV
Contingency: YABUCOA - LUCAS 100.0 kV L
Loading: 151.9% (Rating: 150 MVA)
DFAX: 61.9%
Pre-project loading: 83.5%
2. Facility: YABUCOA SUBSTATION - LUCAS 100.0 kV
Contingency: YABUCOA - SATICOY 100.0 kV Line
Loading: 142.6% (Rating: 111 MVA)
DFAX: 45.4%
Pre-project loading: 75.1%
3. Facility: LUCAS SUBSTATION - HAYFORD 100.0 kV Line
Contingency: YABUCOA - SATICOY 100.0 kV
Loading: 104.5% (Rating: 118 MVA)
DFAX: 45.4%
Pre-project loading: 41.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
YABUCOA - SATICOY 100.0 kV	Rebuild line with higher-capacity conductor	\$1,302,000	42% (\$548,000)
LUCAS - YABUCOA 100.0 kV	Rebuild line with higher-capacity conductor	\$975,000	42% (\$412,000)
LUCAS SUBSTATION - HAYFORD 100.0 kV Line	Reconductor line	\$377,000	42% (\$159,000)
	Total allocated to project		\$1,119,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.